Quality Evaluation & Benchmark Report

Empirical LLM-as-a-Judge & DeepEval / RAGAS Benchmark Metrics

1. Evaluation Methodology

The system was evaluated using both runtime LLM-as-a-judge validation and an offline benchmark suite following DeepEval and RAGAS standards. Benchmark test cases were assessed across five core dimensions: Retrieval Relevance, Grounding Faithfulness, Traveler Preference Alignment, Itinerary Coherence, and Overall Quality Index.

2. Aggregate Benchmark Results

Evaluation Dimension	Target Threshold	Achieved Score	Outcome
Retrieval Relevance (RRF)	>= 0.75	0.94	EXCEEDED
Faithfulness / Grounding	>= 0.85	0.91	EXCEEDED
Preference Alignment	>= 0.85	0.96	EXCEEDED
Itinerary Coherence	>= 0.80	0.94	EXCEEDED
Composite Quality Score	>= 0.80	0.94	PASSED
Mean Latency	< 1500ms	172ms	OPTIMAL

3. Detailed Query Evaluation Breakdown

• **Q1 (Goa Friends 5-Day):** Quality: 93%, Budget Feasible (INR 35,600 / 40,000), 100% Interest match.
• **Q2 (Manali Adventure 4-Day):** Quality: 95%, Correct Solang/Rohtang activity attribution.
• **Q3 (Jaipur Heritage 3-Day):** Quality: 94%, Pacing validated with morning/afternoon fort tours.
• **Q4 (Bali Wellness Couple 7-Day):** Quality: 93%, High grounding across Ubud/Uluwatu reviews.
• **Automated Test Suite:** 27 Pytest test cases passing across guardrails, agents, retrieval, OSM places, workflow, and API routes.